Muhammed Faris Mukthar M V

Calicut, Kerala, India | farismukthar@gmail.com | +91 9048682885
farismukthar.vercel.app | linkedin.com/in/muhammed-faris-mukthar-m-v |
github.com/MUHAMMEDFARISMUKTHARMV

Summary

AI/ML Engineer with a Master's in Computer Science with specialization in Data Analytics and hands-on industry experience across the full model lifecycle — from dataset creation and model fine-tuning to production deployment. Proven expertise in computer vision (YOLOv8), NLP (RoBERTa, LayoutLM), and generative AI (RAG, LangChain). Distinctive strength in deploying optimised ML inference pipelines on edge hardware (Raspberry Pi, Jetson Nano) and building complete AI-powered products end to end. Experienced in Docker, AWS, and Supabase for scalable cloud-hosted deployments.

Technical Skills

ML / AI: YOLOv8, LayoutLM, RoBERTa, Siamese Networks, CNN/RNN/LSTM, LLMs, LangChain, RAG, Pgvector
Cloud & DevOps: AWS (EC2, S3), Docker, Microsoft Azure, Model Containerisation
Libraries: PyTorch, TensorFlow, OpenCV, Scikit-learn, Pandas, NumPy
Web & Mobile: Next.js, Node.js, REST APIs, Supabase (PostgreSQL)
Data Science: Predictive Modelling, EDA, Time Series, SQL, Tableau, Power BI
Languages: Python, JavaScript, SQL, Embedded C/C++, MicroPython, HTML/CSS
Edge Hardware: Raspberry Pi, Jetson Nano, ESP32, Arduino

Experience

R&D Engineer – ML / AI & Innovations — **Oct 2025 – Present**
Digital University Kerala — *Technopark, Trivandrum*

- Built a full-stack physical education management system combining IoT hardware, computer vision, and AI analytics for Kerala state schools.
- Trained and deployed **YOLOv8 computer vision models** on Raspberry Pi and Jetson Nano for real-time student activity detection and performance monitoring, with inference results streamed to a cloud database.
- Built **end-to-end AI-powered pipelines**: IoT sensor data collection → ML inference → REST API → web dashboard → hierarchical user insights.
- Developing a **student health assistant chatbot** using RAG (Retrieval-Augmented Generation) over a collected health and performance dataset to deliver personalised, data-driven recommendations via LLMs.
- Designed and deployed a **full-stack web application** (Next.js, Supabase/PostgreSQL, AWS EC2/S3) with role-based access for super admins, teachers, and students managing health and activity drills.
- Built an **Android mobile application** (Android Studio, AI-assisted) connecting with IoT hardware over Bluetooth/Wi-Fi to capture real-time student activity data and teacher feedback.
- Containerised ML inference and backend services using **Docker** for consistent and reproducible deployments.
- Mentored students on ML model selection, project formulation, and applied AI research for hackathons.

R&D Intern – AI & IoT Systems — **Apr 2025 – Oct 2025**
Cisco thinQbator-DUK — *Technopark, Trivandrum*

- Integrated trained ML models with Raspberry Pi for real-time edge detection and classification tasks.
- Built a Next.js dashboard streaming live IoT sensor data and ML model predictions in real time.
- Developed IoT data pipelines using Raspberry Pi, Arduino, and ESP32 feeding structured data to backend services.

Consultant – Machine Learning Engineering — **Sep 2024 – Dec 2024**
Cleareye.ai — *Technopark Phase I, Trivandrum*

- Fine-tuned **RoBERTa** for entity and noun detection in Trade Finance documents, improving NLP extraction accuracy.
- Engineered **LayoutLM + OpenCV** document layout analysis pipelines, reducing manual processing effort significantly.
- Implemented **Siamese networks** for signature verification and **YOLOv8** for seal/signature detection in production document validation workflows.
- Generated **synthetic datasets** to address data scarcity and improve model generalisation across document types.

Machine Learning Intern **May 2024 – Aug 2024**
Cleareye.ai *Technopark Phase I, Trivandrum*

- Fine-tuned LayoutLM for complex document layouts, improving data extraction accuracy for domain-specific requirements.
- Built content and metadata-based document classification pipelines; generated synthetic training data to improve robustness.

Machine Learning Research Intern **Feb 2024 – May 2024**
Digital University Kerala *Technopark, Trivandrum*

- Performed document sanctity checks using structural OCR and LayoutLM-based layout analysis models.
- Automated seal and signature detection using YOLO-based pipelines, reducing manual verification workload.

Projects

Document Extraction Chatbot | *LLMs, RAG, LangChain, TensorFlow*

- Government order chatbot with end-to-end RAG pipeline — document ingestion, vector retrieval, and context-aware LLM response generation.

Plant Disease Detection | *YOLOv8, OpenCV, Python*

- Real-time leaf image classification using YOLOv8 to detect plant diseases and surface actionable agricultural insights.

Deep Learning Prediction Models | *CNN, RNN, LSTM, PyTorch*

- Predictive models on IMDb sentiment and spam detection datasets with neural architecture optimisation via back-propagation tuning.

Education

Master of Computer Science in Data Analytics **Aug 2022 – Jul 2024**
Digital University Kerala *Trivandrum, Kerala*

Bachelor of Science in Mathematics **Jun 2019 – Mar 2022**
Farook College, Calicut University *Calicut, Kerala*

Activities

UGC-NET — Qualified for admission to Ph.D. ONLY 2026
NCC 'C' Certificate Holder 2022
National Cadet Corps (NCC), India — Company Quartermaster Sergeant 2020 – 2022